# CreditPathAI – Feature Engineering Handout (Loan Default Prediction)

This handout covers Feature Engineering, the most critical stage in improving model performance. After completing EDA, we now transform insights into meaningful features that help machine learning models better understand borrower risk. Feature engineering is what separates a basic project from an industry-level system.

## Step 1 – Load Clean Dataset

import pandas as pd df = pd.read_csv("clean_loans.csv")

## Step 2 – Loan to Income Ratio

df['loan_income_ratio'] = df['loan_amount'] / df['income'] Reason: High loan compared to income → higher default risk

## Step 3 – Credit Utilization Proxy

df['credit_utilization'] = df['loan_amount'] / df['num_credit_lines'] Reason: More debt per credit line → higher risk

## Step 4 – Interest Burden

df['interest_burden'] = df['loan_amount'] * df['interest_rate'] Reason: Higher interest → repayment difficulty

## Step 5 – Employment Stability

df['employment_stability'] = df['months_employed'] / df['age'] Reason: Stable employment → lower risk

## Step 6 – Debt to Income Flag

df['high_dti_flag'] = (df['dti_ratio'] > 0.5).astype(int) Reason: High DTI strongly correlates with default

## Step 7 – Credit Score Bucketing

df['credit_score_bucket'] = pd.cut(df['credit_score'], bins=[0,600,700,800,900], labels=['Poor','Average','Good','Excellent']) df = pd.get_dummies(df, columns=['credit_score_bucket']) Reason: Categorizing credit scores improves model learning

## Step 8 – Log Transform Income

import numpy as np df['log_income'] = np.log1p(df['income']) Reason: Income is usually skewed → log stabilizes distribution

## Step 9 – Interaction Feature

df['income_loan_interaction'] = df['income'] * df['loan_amount'] Reason: Captures combined financial pressure

## Step 10 – Drop Weak Features

df = df.drop(columns=['income']) # example if redundant Reason: Remove noisy or redundant features

## Step 11 – Final Dataset Save

df.to_csv("final_features.csv", index=False)

## Outcome

Students now have a feature-rich dataset ready for: • Logistic Regression • XGBoost • LightGBM Better features → better predictions.